



API Documentation

QR API with Expiry

Contents

Introduction 2

API Details 3

1. API Name: QR API 4

2. API Name: Callback 8

3. API Name: Transaction Status 10

4. API Name: Callback Status 13

5. API Name: Refund API 17

6. Error Codes 21

Security 31

Encryption & Decryption Process 31

Introduction:

UPI is a set of APIs created by NPCI to facilitate online immediate payments. UPI is expected to further propel easy instant payments via mobile. The payments can be both sender (payer) and receiver (payee) initiated and can be carried out using virtual payment addresses, Aadhaar integration, mobile number etc. The payer's smartphone could be used for secure credential capture.

Merchant on-boarding:

Merchant needs to provide the following information for onboarding of UAT and production environment:

Technical list:

- IP address (For dynamic IPs please provide range of IP addresses)
- Merchant call-back URL to post final transaction status from ICICI's end
- Merchant certificate with 4096 bits public key (.pem or .cer format) for encryption ○
Merchant SSL certificate for sending call back response on call back url

Once the merchant provides all the above-mentioned technical list, Bank will do the necessary configuration at their end and provide Merchant ID (MID) which shall be configured against the Virtual Payment Address (VPA). Once these details are received at merchant's end, they can start the API testing.

Bank will also provide ICICI bank's public key certificate for encryption to be done at merchant's end. Merchant will need to make encrypted request call using ICICI Bank's public key certificate to selected APIs from their Application Server and ICICI Bank will post encrypted response packet using merchant's public key certificate. Merchant is required to decrypt the response packet received at their end with the corresponding private key.

General Flow:

1. For QR code or intent call transactions, merchant needs to send 'refid' in "tr" field of the QR/intent string.
2. There are two ways to generate 'refid'.
 - First, is to call QR API. Merchant will send QR API request to QR API. On receiving request in correct format, ICICI bank will send QR API response which will have unique Refid starting with EZY or EZP prefix.
 - Second way, merchant will generate its own 'refid' with merchant specific three Prefix letter.
3. Using 'refid' merchant can create QR code or can initiate intent call. Customer will scan QR code or in case of intent call, customer will choose his PSP app to complete transaction.
4. Once Customer accepts or rejects the request from his mobile app, ICICI bank will send call back response to merchant stating Success or Reject on their callback URL.
5. When customer accepts the request from his mobile, transaction will be completed, and amount will be credited to merchant's account.

6. For 'refid' generated through QR API ICICI bank will send respective 'merchantTranId' of QR API request in call back response.
7. For merchant generated 'refid', ICICI bank will send merchant generated 'refid' in call back response except prefix letter.

API Details:

The specific name of each APIs are mentioned in the below sections. The customer parameters to be passed are specific to each API.

Below is the format for sending details.

First the parameters and their values will be entered in JSON Object. Then the whole JSON object will be encrypted and then encoded. Finally, the whole request will be passed through URL.

It will be a POST request.

[GatewayURL(Base64Encode(RSA_Encrypt(JSON_Object{Field_Elements(field1,field2,...)})))]

The JSON Request Object is mentioned below where complete payload is encrypted using the public key provided by ICICI Bank: {

```
Base64Encode(RSA_Encrypt( { "merchantId" : "111111", "subMerchantId" : "12234",
"terminalId" : "2342342", "merchantTranId" : "612413726581",...,... }))) }
```

Encryption needs to be done using RSA 4,096 bits public key provided by ICICI Bank.

While sending the request please add the Headers in CODE which are **Highlighted**:

```
accept:*/*, accept-encoding:*, accept-language:en-US,en;q=0.8,hi;q=0.6, cache-control:no-cache,
connection:keep-alive, content-length:684, content-type:text/plain;charset=UTF-8,
host:apigwuat.icicibank.com:8443, origin:chrome-extension://fhhbjgbiflinjbdggehcdcbncdddomop,
postmantoken:bfd89d8e-fd90-b9b7-b9da-17469eb99976, user-agent:Mozilla/5.0 (Windows NT 6.3; WOW64)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/54.0.2840.71 Safari/537.36.
```

Below steps are to be followed:

- 1) Use Public certificate of ICICI Bank to encrypt payload using Algorithm RSA/ECB/PKCS1Padding Called Encrypted Payload
- 2) Base64 encode the Encrypted payload to get EncodedEncryptedPayload
- 3) The final string generated is the Request to API Gateway
- 4) Use content-type as Plaint/text in the body and call the API

Sample Payload:

```
X30i3+Y5kWiuQQ6/d+pW6oJaMidDpaXLznH03XUm6xRIUeAhKTghFb2SeXHzyNCkoi2+Ci8ms2OU
ljUhsJTyLWo+N6INqMNPki3ieQWBAXo+8s/xc9t/SSp3eLUIPgcEnwHJ93tDnvzD8KjRtWqo3mBg
ja84TnQvISM918WcUvZQi/NLbGjxlemBm2bHJYSfJwVtTmBjUvblZmAhrW14YpfY6B8ZzUBujZhf
qldMLL+B+zyKd9tITztVCeVINQvDPhsNnU9OBNN+sHIESZzzi+B7PgYn7n/Mzpo594npZbZ9sDwS
XdwMIK1KY3rJXfzoq+RZL+dc1ftfZjlqCHFfHposHzB3C3Smjm9EnzZEB0DfnxnT5CHvWM8I9OI
1CKew9ZjKrbHAQ6y1eKDKad9935TISh/WTirdtDHcpJW+HC8NYzd5lwzuVvWr24JD+riS7DcnKv7
YDH4xxnHaWXx/g8tgsxWK4H2m+VdvivVKWzAaX4GeNZd76uxGGUvKwxgqiyLasFqtYzYjOI8fRq
jGpDdQ9CkrdmvyOOdOV+qFbXaMxCLyBAIrarFGO4QWoO5oJvWY6zOXa/A2Apx+IX7CG51VuiwQZ
ssVEAGVzHQYn1n69nf2Jj/LLJXbg9gFg9naHTwf2m9jBorUhoo007Cm87v5oytwzGJIX13VOIAY=
```

1. API Name: QR API (with expiry)

Description: QR API will be used to fetch 'refId' from ICICI system. This 'refId' will be used to generate QR/intent string.

As an optional feature, Validation of the Debit Account can be done using Validate Payer Acc Flag, Payer Account and Payer IFSC parameter.

To introduce the feature of expiry in the QR, additional input fields are added, namely validityStartDateTime and validityEndDateTime. The QR can be made a time bound QR by using the these tags.

The validity of the QR can also be updated by using the "update" tag . This can be done anytime before the expiry or before payment on the QR, whichever earlier.

End Points:

UPI 1.0 End Points

UAT Endpoint:

<https://apibankingonesandbox.icici.bank.in/api/MerchantAPI/UPI/v0/QR3/{merchantId}>

Live Endpoint: <https://apibankingone.icici.bank.in/api/MerchantAPI/UPI/v0/QR3/{merchantId}>

UPI 2.0 End Points

UAT Endpoint:

<https://apibankingonesandbox.icici.bank.in/api/MerchantAPI/UPI2/v1/QR>

Live Endpoint: <https://apibankingone.icici.bank.in/api/MerchantAPI/UPI2/v1/QR>

Input Parameters:

Name	Type	Description	Mandatory (Y-Yes / O- Optional/Conditional)	Length
merchantId	Number	Merchant Identification Number	Y	10

terminalId	Number	Needs to send Merchant category code (MCC code).	Y	4
amount	Number	Amount to be debited (In Rupees) in Integer value with 2 decimal E.g. : '200.00' or '300.12'	Y	20
merchantTranId	AlphaNumeric	This will be a Transaction ID generated by the API and should always be unique	Y	35
billNumber	AlphaNumeric	Bill Number / Order Number	Y	50
validatePayerAccFlag	String	'Y' for validating debit a/c details or 'N' for nonvalidation	O	1
payerAccount	Number	When 'validatePayerAcc' Flag is 'Y' then it is mandatory. Payer Account number is required to be entered.	O	
payerIFSC	AlphaNumeric	When 'validatePayerAcc' Flag is 'Y' then it is mandatory. Payer IFSC code is required to be entered.	O	
splitName	String	Type of split or convenience fee. "CCONFEE" and "PCONFEE"	O	25
splitValue	Number	Amount of the split or convenience fee.	O	20

refId	Alphanumeric	Reference id that has to be updated. This has to be passed only when an existing QR has to be updated. The update tag should also be Y in this case.	C	
validityStartDateTime	Date Time	Start date of QR validity period (Ex: DD/MM/YYYY HH:MM:SS)	O	
validityEndDateTime	Date Time	End date of QR validity period (Ex: DD/MM/YYYY HH:MM:SS)	O	
update	String	'Y' for updating an existing QR or 'N' for creation of new QR	O	

Sample Request Packet for QR creation:

```
{
  "amount": "5.00",
  "merchantId": "118449",
  "terminalId": "5411",
  "merchantTranId": "p0nillp0k9lqlp091p17",
  "billNumber": "sdf1po111b",
  "validatePayerAccFlag": "Y",
  "payerAccount": "0405012740",
  "payerIFSC": "IC00000",
  "refId": ""
  "validityStartDateTime": "29/12/2022 15:30:00",
  "validityEndDateTime": "29/12/2022 16:30:00",
  "update": "N"
}
```

If no value is passed in validity end date parameter **default expiry of 30 days** will be considered.

Sample Request Packet for updating the expiry:

```
{
  "amount": "5.00",
  "merchantId": "118449",
  "terminalId": "5411",
```

```

"merchantTranId": "p0nillp0k9lqlp091p17",
"billNumber": "sdf1po111b",
"validatePayerAccFlag": "Y",
"payerAccount": "0405012740",
"payerIFSC": "IC00000",
"refid": "EZY286844327832"
"validityStartDateTime": "29/12/2022 15:40:00",
"validityEndDateTime": "29/12/2022 16:30:00",
"update": "Y"
}

```

Output Parameters:

Name	Type	Description
merchantId	Number	Merchant Identification Number
terminalId	Number	Merchant category code
success	Alphanumeric	Result of the API Call
response	String	Response code indicating status of QR API
message	String	message description indicating status of QR API
merchantTranId	Alphanumeric	This will be a Unique Transaction ID generated by the Merchant.
refid	Alphanumeric	Reference id

Sample Response Packet for QR generation:

```

{
"response": "0",
"merchantId": "118449",
"terminalId": "5411",
"success": "true",
"message": "Transaction Initiated",

```



```
"merchantTranId": "p0nillp0k9lqlp091p17",
"refId": "EZY286844327832",
}
```

Sample Response Packet for updating the expiry:

```
{
"response": "0",
"merchantId": "118449",
"terminalId": "5411",
"success": "true",
"message": "Transaction Initiated",
"merchantTranId": "p0nillp0k9lqlp091p17",
"refId": "EZY286844327832",
}
```

QR code generation string:

upi://pay?pa=<merchant VPA>&pn=<merchant name>&tr=<Refid>&am=<amount>&cu=INR&mc=<MCC code>

upi://pay?pa=<merchant VPA>&tr=<Refid generated via QR API/Ref id generated by Merchant>&am=<amount>&cu=INR&mc=<MCC code>&split=Split Name:Split Value

Example: upi://pay?pa=abc@icici&pn=Abc&tr= EZY286844327832&am=5.00&cu=INR&mc=5411

Note: For QR type transactions with **MIDs procured through DMO services**. The merchantTranId will be the same value as the refid passed inside the “tr” field of QR string ie. Prefix will be present in the merchantTranId received in callback for Merchant generated RefID.

2. API Name: Callback

Description: Final transaction response posted by ICICI Bank to Merchant’s callback URL.

Parameters:

Name	Type	Description	Length
merchantId	Number	Merchant Identification Number	10
subMerchantId	Number	Sub Merchant Identification Number of Merchant	10

terminalId	Number	Merchant category code (MCC code).	10
------------	--------	------------------------------------	----

BankRRN	Number	Bank reference Number for this transaction	20
merchantTranId	AlphaNumeric	Transaction Id as sent in the request packet	35
PayerName	AlphaNumeric	Name of the Payer	50
PayerMobile	Number	Mobile Number of the Payer	10
PayerVA	AlphaNumeric	Virtual Payment Address of the Payer	255
PayerAmount	Numeric	Amount of the Transaction	20
TxnStatus	AlphaNumeric	Status of the Transaction	20
TxnInitDate	Date and Time	Date and time on which the transaction was initiated	20
TxnCompletionDate	Date and Time	Date and time on which the transaction was completed	20
ResponseCode	AlphaNumeric	The response code received from NPCI	50

RespCodeDescription	String, Special characters	The exact reason of failure in the format: "response code description error code error code description" as received from NPCI	255
PayeeVPA	AlphaNumeric, Special Characters	Virtual Payment Address of the Payee	250
PayerAccountType	AlphaNumeric, Special Characters	Account Type of the Payer. Type of Accounts - Savings, Current , Default, NRE, NRO , CREDIT , PPI WALLET , SOD , UOD In case of CREDIT txn, BIN data is also shared in the below format. Possible Values: "CREDIT 0000465278" or "Savings" BIN is shared only for Credit Card Txn's	50
SplitName	Alphabetic	Type of split or convenience fee. "CCONFEE" and "PCONFEE"	25
SplitValue	Number	Amount of the split or convenience fee.	20

Sample Callback: Successful transaction Callback response after decryption.

```
{
  "merchantId" : "401491",
  "subMerchantId" : "401491",
  "terminalId" : "5411",
  "BankRRN" : "615519221396",
  "merchantTranId" : "612411454593",
  "PayerName" : "hhjjj",
  "PayerMobile" : "8879770059",
  "PayerVA" : "testing1@imobile",
  "PayerAmount" : "12",
```

```

"TxnStatus" : "SUCCESS",
"TxnInitDate" : "201904085142352",
"TxnCompletionDate" : "20190408142352"
"ResponseCode" : "00",
"RespCodeDescription": "APPROVED OR COMPLETED SUCCESSFULLY "
"PayeeVPA": "fsinsurancesit2@icici"
"PayerAccountType": "Savings"
"SplitName": "CCONFEE",
"SplitValue": "1.00" }

```

Sample Callback: Fail transaction Callback response after decryption.

```

{
  "merchantId" : "401491",
  "subMerchantId" : "401491",
  "terminalId" : "5411",
  "BankRRN" : "615519221396",
  "merchantTranId" : "612411454593",
  "PayerName" : "hhjjj",
  "PayerMobile" : "8879770059",
  "PayerVA" : "testing1@imobile",
  "PayerAmount" : "12",
  "TxnStatus" : "FAILURE",
  "TxnInitDate" : "201904085142352",
  "TxnCompletionDate" : "20190408142352"
  "ResponseCode" : "U19",
  "RespCodeDescription" : "REQUEST AUTHORISATION IS
    DECLINED|ZA|TRANSACTION DECLINED BY CUSTOMER" ,
  "PayeeVPA" : "fsinsurancesit2@icici"
  "SplitName": "CCONFEE",
  "SplitValue": "1.00"
}

```

The three tags in the above sample callback (**ResponseCode** , **RespCodeDescription** and **PayeeVPA**) are optional to the merchants, i.e. they'll be present in the Callback only if the merchant has enabled the tags.

For CC on UPI, the PayerAccountType will be "CREDIT|0000465278"

For Asymmetric encryption enabled MIDs, Callbacks will fail if Split Tags are sent as the packet size will increase. Hence for Merchants to use Split Tags, Hybrid Encryption is mandatory.

3. API Name: Transaction Status

Description: This API will be used by Merchant to get the status of the transaction based on 'merchantTranID' input parameter.

UPI1.0 Endpoints

UAT Endpoint:

<https://apibankingonesandbox.icici.bank.in/api/MerchantAPI/UPI/v0/TransactionStatus3/{merchantId}>

Live Endpoint: <https://apibankingone.icici.bank.in/api/MerchantAPI/UPI/v0/TransactionStatus3/{merchantId}>

UPI2.0 Endpoints

UAT Endpoint: <https://apibankingonesandbox.icici.bank.in/api/MerchantAPI/UPI2/v1/TransactionStatus>

Live Endpoint: <https://apibankingone.icici.bank.in/api/MerchantAPI/UPI2/v1/TransactionStatus>

Input Parameters:

Name	Type	Description	Mandatory (Y/N)	Length
merchantId	Number	Merchant Identification Number	Y	10
subMerchantId	Number	Sub Merchant Identification Number of Merchant	Y	10
terminalId	Number	Needs to send Merchant category code (MCC code)	Y	4
merchantTranId	AlphaNumeric	This will be a Transaction ID generated at the time of original request	Y	35

Sample Packet

```
{  
  "merchantId": "118449",  
  "subMerchantId": "118449",  
  "terminalId": "5411",  
  "merchantTranId": "p0nillp0k9lqlp091p17"  
}
```

Output Parameters:

Name		Type	Description
Response	Number		Response Code
merchantID	Number		Merchant Identification Number
subMerchantId	Number		Sub Merchant Identification Number of Merchant
terminalId	Number		MCC
Success	String		Result of the API Call
Message	String		Response Code Description
merchantTranId	AlphaNumeric		This will be a Unique Transaction ID generated by the Merchant.
OriginalBankRRN	Number		Reference Number generated by Bank
Amount	Number		Amount
Status	AlphaNumeric		Status of the transaction
respCodeDesription	String, Special characters		The exact reason of failure in the format: " response code description error code error code description" as received from NPCI
splitName	Alphabetic		Type of split or convenience fee. "CCONFEE" and " PCONFEE"

splitValue	Number	
		Amount of the split or convenience fee.

Sample Response: Successful transaction response after decryption.

```
{
  "response" : "0",
  "merchantId" : "106161",
  "subMerchantId" : "12234",
  "terminalId" : "5411",
  "OriginalBankRRN" : "615519221396",
  "PayerVA" : "testing1@imobile",
  "amount" : "12",
  "success" : "true",
  "message" : "Transaction Successful",
  "status" : "SUCCESS"
  "respCodeDescription" : " APPROVED OR COMPLETED SUCCESSFULLY "
  "splitName": "CCONFEE",
  "splitValue": "1.00"
}
```

Sample Response: Fail transaction response after decryption.

```
{

  "response" : "0",
  "merchantId" : "106161",
  "subMerchantId" : "12234",
  "terminalId" : "5411",
  "OriginalBankRRN" : "214700801753",
  "PayerVA" : "testing1@imobile",
  "amount" : "12",
  "success" : "true",
  "message" : "Transaction Successful",
  "status" : "FAIL"
  "respCodeDescription" : " U31 | CREDIT HAS BEEN FAILED|ZY|INACTIVE OR DORMANT ACCOUNT
(BENEFICIARY)"
  "splitName": "CCONFEE",
  "splitValue": "1.00"
}
```

The tag (**respCodeDescription**) in the above sample response is optional to the merchants, i.e. it will be present in the response only if the merchant has enabled the tag. Please note that the tag will be visible only if UPI 2.0 services are being used.

Current response:
PENDING, SUCCESS, FAILURE

4. API Name: Callback Status

Description: This API will be used by Merchant to get the status of the transaction by passing correct transaction type.

UPI 1.0 End Points

UAT Endpoint: <https://apibankingonesandbox.icici.bank.in/api/MerchantAPI/UPI/v0/CallbackStatus2/{merchantId}>

Live Endpoint: <https://apibankingone.icici.bank.in/api/MerchantAPI/UPI/v0/CallbackStatus2/{merchantId}>

UPI 2.0 End Points

UAT Endpoint:
<https://apibankingonesandbox.icici.bank.in/api/MerchantAPI/UPI2/v1/TransactionStatusByCriteria>

Live Endpoint:
<https://apibankingone.icici.bank.in/api/MerchantAPI/UPI2/v1/TransactionStatusByCriteria>

Input Parameters:

Name	Type	Description	Mandatory (Y/C)	Length
merchantId	Number	Merchant Identification Number	Y	10
subMerchantId	Number	Sub Merchant Identification Number of Merchant	Y	10
terminalId	Number	Needs to send Merchant category code (MCC code)	Y	4

transactionType	Alphabet	Flag to identify type of original transaction as C, R, Q or P as per below mentioned table*	Y	1
merchantTranId	AlphaNumeric	This will be a Transaction ID generated at the time of original request.	C	35
transactionDate	Date	Date of the Transaction	C	20
BankRRN	Number	Bank Reference Number of the original transaction	C	15
refID	AlphaNumeric	Reference Number passed in QR/Intent Call	C	

*Transaction Type Flag	Scenario	MerchantT ranId	BankRRN	RefId	Transaction Date
C	Collect Pay Transaction	Mandatory	Non-Mandatory	N/A	N/A
R	Refund Transactions	Mandatory	Non-Mandatory	N/A	N/A
Q	ICICI generated ref-Id (EZY or EZP)	Mandatory	Non-Mandatory	Non-Mandatory	N/A
P	Merchant generated refId or Push	Any 1 out of 3	Any 1 out of 3	Any 1 out of 3	Mandatory when search based on Ref Id

Sample Packet:

```
{
  "merchantId": "118449",
  "subMerchantId": "118449",
  "terminalId": "5411",
  "transactionType": "C",
  "merchantTranId": "p0nillp0k9lqlp091p17" }
```

Output Parameters:

Name	Type	Description
Response	Number	Response Code
merchantId	Number	Merchant Identification Number
subMerchantId	Number	Sub Merchant Identification Number of Merchant
terminalId	Number	MCC
success	String	Result of the API Call
message	String	Response Code Description

merchantTranId	AlphaNumeric	This will be a Unique Transaction ID generated by the Merchant.
OriginalBankRRN	Number	Reference Number generated by Bank
PayerVA	AlphaNumeric	Virtual Payment Address of the Payer
Amount	Number	Amount
status	AlphaNumeric	Status of the transaction
TxnInitDate	Date and Time	Date and time on which the transaction was initiated
TxnCompletionDate	Date and Time	Date and time on which the transaction was completed
respCodeDescription	String, Special characters	The exact reason of failure in the format: “ response code description error code error code description” as received from NPCI

refundRRN	Number	Reference number generated by Bank for refunds. This will only come for Status check request where transactionType "R".
splitName	Alphabetic	Type of split or convenience fee. "CCONFEE" and " PCONFEE"
spiltValue	Number	Amount of the split or convenience fee.

Sample Response: Successful transaction response after decryption.

```
{
  "response" : "0",
  "merchantId" : "106161",
  "subMerchantId" : "12234",
  "terminalId" : "5411",
  "OriginalBankRRN" : "615519221396",
  "merchantTranId" : "612411454593",
  "Amount" : "12",
  "payerVA" : " testing1@imobile ",
  "success" : "true",
  "message" : "Transaction Successful",
  "status" : "SUCCESS",
  "TxnInitDate" : "20160715142352",
  "TxnCompletionDate" : "20160715142352"
  "respCodeDescription" : "APPROVED OR COMPLETED SUCCESSFULLY "
  "splitName": "CCONFEE",
  "splitValue": "1.00"
}
```

Sample Response: Fail transaction response after decryption.

```
{
  "response" : "0",
```

```

"merchantId" : "106161",
"subMerchantId" : "12234",
"terminalId" : "5411",
"OriginalBankRRN" : "214700801759",
"merchantTranId" : "612411454593",
"Amount" : "12",
"payerVA" : " testing1@imobile ",
"success" : "true",
"message" : "Transaction Successful",
"status" : "FAIL",
"TxnInitDate" : "20160715142352",
"TxnCompletionDate" : "20160715142352"
"respCodeDescription" : "REQUEST AUTHORISATION IS DECLINED|ZA|TRANSACTION DECLINED
BY CUSTOMER"
"splitName": "CCONFEE",
"splitValue": "1.00"

}

```

The tag (**respCodeDescription**) in the above sample response is optional to the merchants, i.e it will be present in the response only if the merchant has enabled the tag. Please note that the tag will be visible only if UPI 2.0 services are being used.

5. API Name: Refund API

Description: This API needs to be used by Merchants to initiate refunds of the transactions. Both offline and online refunds are supported in the same API.

UPI 1.0 End Points

UAT Endpoint: <https://apibankingonesandbox.icici.bank.in/api/MerchantAPI/UPI/v0/Refund2/{merchantId}>

Live Endpoint: <https://apibankingone.icici.bank.in/api/MerchantAPI/UPI/v0/Refund2/{merchantId}>

UPI 2.0 End Points

UAT Endpoint: <https://apibankingonesandbox.icici.bank.in/api/MerchantAPI/UPI2/v1/Refund>

Live Endpoint: <https://apibankingone.icici.bank.in/api/MerchantAPI/UPI2/v1/Refund>

Input Parameters:

Name	Type	Description	Mandatory (Y/N)	Length
merchantId	Number	Merchant Identification Number	Y	10
subMerchantId	Number	Sub Merchant Identification Number of Merchant	Y	10
terminalId	Number	Needs to send Merchant category code (MCC code).	Y	4
originalBankRRN	String	Original Transaction Id	Y	15
merchantTranId	String	Refund Transaction Id	Y	35
originalMerchantTranId	AlphaNumeric	Merchant TranID of Refund transaction.	Y	35
refundAmount	Number	Amount to be debited. (In Rupees, Integer value with 2 decimal)E.g. : 200.00 / 300.12	Y	20
payeeVA	AlphaNumeric	Alias name with which the payee can be identified by his registered entity.	N	255
Note	AlphaNumeric	Remarks entered by the payer for his reference.	Y	50

onlineRefund	String	Refund request mode – Online or Offline refund – ‘Y’ for online refund and ‘N’ for Offline refund	Y	1
--------------	--------	---	---	---

Sample Packet:

```
{
  "merchantId": "106092",
  "subMerchantId": "12234",
  "terminalId": "2342342",
  "originalBankRRN": "622415338172",
  "merchantTranId": "88442047",
  "originalmerchantTranId": "202020202021",
  "payeeVA": "yatin@imobile",
  "refundAmount": "10.00",
  "note": "refund-request",
  "onlineRefund": "Y"
}
```

Output Parameters:

Name	Type	Description
Response	Number	Response Code
merchantId	Number	Merchant Identification Number
subMerchantId	Number	Sub Merchant Identification Number of Merchant

terminalId	Number	MCC
success	String	Result of the API Call
message	String	Response Code Description
merchantTranId	AlphaNumeric	This will be a Unique Transaction ID generated by the Merchant.
OriginalBankRRN	Number	Reference Number generated by Bank. For Online refund, new RRN will be generated. For Offline, Original RRN will be returned
status	AlphaNumeric	Status of the transaction
respCodeDescription	String	The exact reason of failure in the format: “ response code response code description error code error code description” as received from NPCI

Sample Response

```
{
  "merchantId": "106092",
  "subMerchantId": "12234",
  "terminalId": "2342342",
  "success": "true",      "response": "0",
  "status": "SUCCESS",
  "message": "Transaction Successful",
  "originalBankRRN": "622415338172",
  "merchantTranId": "88442055"
  "respCodeDescription": "APPROVED OR COMPLETED SUCCESSFULLY"
```



```
}
```

Sample Response:

1) Online Refund:

Success Response:

```
{
  "merchantId": "106092",
  "subMerchantId": "12234",
  "terminalId": "2342342",
  "success": "true",
  "Amount": "10.00",
  "response": "0",
  "status": "SUCCESS",
  "message": "Transaction Successful",
  "originalBankRRN": "622415338172",
  "merchantTranId": "88442055"
  "respCodeDescription" : "APPROVED OR COMPLETED SUCCESSFULLY"
}
```

Failure Response:

```
{
  "subMerchantId": 400899,
  "merchantId": 400668,
  "success": false,
  "response": "U31",
  "Amount": "10.00",
  "originalBankRRN": 225600934895,
  "terminalId": 5411,
  "message": "U31 | Credit has been failed",
  "merchantTranId": "88442055",
  "status": "FAILURE"
  "respCodeDescription": "U31 | CREDIT HAS BEEN FAILED | B3 | TRANSACTION NOT PERMITTED TO THE
ACCOUNT (EXAMPLE: MINOR ACCOUNT, PROPRIETOR ACCOUNT, LEGAL CASE AGAINST THIS ACCOUNT
ETC., NRE (AS PER BANK'S POLICY))",
}
```

2) Offline Refund:

```
{
  "merchantId": "106092",
  "subMerchantId": "106092",
  "terminalId": "5414",
  "success": "true",
  "Amount": "10.00",
  "response": "0",
  "status": "SUCCESS",
  "message": "Refund transaction initiated",
  "originalBankRRN": "622415338172",
}
```

```
"merchantTranId":"88442047"
}
```

The tag (**respCodeDescription**) in the above sample response is optional to the merchants, i.e. it will be present in the response only if the merchant has enabled the tag.

Please note that the tag will be visible only if UPI 2.0 services are being used. The URLs refer to the same service.

6. Error Codes

Code	Description	Reasons
500	Internal Server Error	Internal Server Error
401	Unauthorized	APIkey,IP whitelisting or SSL not present

403	Forbidden	Request not proper.
429	Too Many Requests	Too Many Requests
8002	INVALID_JSON.	INVALID_JSON.
8003	INVALID_FIELD FORMAT OR LENGTH	Field is not in the format mentioned
8004	MISSING_REQUIRED_FIELD	Mandatory field is missing
8006	INVALID_FIELD_LENGTH	Length of field exceeds defined length

8007	Invalid JSON, OPEN CURLY BRACE MISSING	Open Brace missing in JSON
8008	Invalid JSON,END CURLY BRACE MISSING	Closing Brace missing in JSON

8009	Internal Server Error	White space characters
------	-----------------------	------------------------

8010	INTERNAL_SERVICE_FAILURE	The system had an internal exception
8011	BACKEND_HOST_NOT_FOUND	The Server referenced in the URL cannot be reached.
8012	BACKEND_CONNECTION_TIMEOUT	Cannot connect to service
8013	BACKEND_READ_TIMEOUT	Cannot read from service
8014	BACKEND_BAD_URL	The URL is incorrect.
8017	INVALID JSON	Improper JSON
8016	Decryption Fail	Request not properly Encrypted
5000	Invalid Request	if the request is failed with some other reasons
5001	Invalid Merchant ID	If the merchant Id is not valid

5002	Duplicate Merchant TranId	Transaction is already initiated with merchant transaction id
5003	Merchant Transaction Id is mandatory	If merchant transaction id null
5004	Invalid Data	invalid packet
5005	Collect By date should be greater than or equal to Current date	If given collect by date is less than current date
5006	Merchant TranId is not available	No transaction initiated with given transaction id based on merchant id

5007	Virtual address not present	If merchant entered invalid customer VPA
5008	PSP is not registered	If merchant entered wrong PSP handler
5009	Service unavailable. Please try later.	Default error response for unexpected internal failures.
5010	Technical Error	If any technical error.

5011	Duplicate merchant TranId	Transaction is already initiated with merchant transaction id
5012	Request has already been initiated for this transaction	If request is initiated already for this transaction.
5013	Invalid VPA	If VPA does not exists
5014	Insufficient amount	If Original amount is less than refund amount
5015	Invalid Original TranId	If original transaction Id is not available
5016	Payee VA should not be Merchant VA	Should not be Merchant Virtual Address

5017	Sorry you can't initiate refund request	Merchant can initiate online refund only if online refund flag is enabled
5018	Merchant VPA and Reference ID is not match	
5019	Invalid Terminal Id	

5020	No response from Beneficiary Bank. Please wait for recon before initiating the transaction again.	For Deemed approved transactions or timed out requests
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5021	Transaction Timed out. Please check transaction status before initiating again.	OSB Timed out for collect request
5022	Terminal Id is mandatory	
5023	Multiple transactions against given parameter. Please provide bank RRN	
5024	Record not found against given parameters	
5025	Please enter valid refund amount	
5026	Invalid Consumer number	
5027	Invalid merchant prefix	

5028	Virtual Address Already Exists	
5029	No Response From Switch	
5030	Please try again	In case Check VPA return actCode 950 from Switch
5031	Validity start date should not be less than current date	If Validity start date is less than current date

5032	Validity end date should not be less than validity start date	If Validity end date is less than validity start date
5033	Mandate request not created	Without initiating the manage mandate
5034	No Approved Mandates are available	If manage mandate request are not in SUCCESS state
5035	Mandate expired	If mandate validity period is completed

5036	Mandate amounts mis-matched	If manage mandate is EXACT and different amount given in execute mandate
------	-----------------------------	--

5037	Execution amount exceeded to Mandate approved amount	If manage mandate is MAX and execution amount crossed in execute mandate
5038	Invalid Validate Payer Account Flag	If validate payer account flag is other than Y and N
5039	Invalid Payer Account	If Payer Account is null, empty or invalid pattern
5040	Invalid Payer IFSC	If Payer IFSC is null, empty or invalid pattern
5041	Invalid Sequence Number	
5042	Duplicate Sequence Number	
5043	Invalid Unique Merchant ReferenceId	

5044	Invalid Merchant Name	
5045	Invalid Marketing Name	
5046	Invalid Bank Assigned MerchantId	
5047	OSB Timeout	
5048	New Unique Merchant Referenceld	
5049	Failed at switch. Please try registering again.	

5050	Details of Bank Assigned MerchantId not found	
5051	Duplicate Unique Merchant Referenceld	

5052	Invalid Mobile Number	
5053	Invalid Action	
5054	Invalid MCC	

5073	Revoke Mandate block should always be N	
5074	Mandate amount and date is same as per old mandate	
5075	Mandate Details not found	
5076	Payer Account Validation Failure	
5077	Transaction Deemed	

5078	Merchant is already registered	
5079	Transaction Unsuccessful.	Same VPA is used by some other Merchant. Kindly register using a new VPA.
5080	Invalid Parent MerchantID	
5081	Invalid Account Number	
5082	Invalid Merchant Legal Name	
5083	Invalid Merchant Business Name	
5084	Invalid IFSC	
5085	Request Timed out. Please try again.	

5087	Payer Account Number and Payer IFSC are mandatory	
------	---	--

5088	OTP is mandatory	
5089	Invalid OTP	
5090	Voucher Validation not Initiated	
5091	Invalid UMN/UUID and MerchantId combination	
5093	Invalid Voucher	
5094	Invalid Mode OR Purpose Code	
5095	Voucher Validation Timed out	

5096	Voucher Redemption Timed out	
------	------------------------------	--

5097	Transaction Failed	
------	--------------------	--

5098	UMN is mandatory for QR based request	
5101	Merchant is Already Registered with this Same PanNumber	Same Aggregator cannot register more than 1 merchant using the same PAN Number.
5102	USED MCC CODE IS RESTRICTED ONE	Some MCCs are not allowed to be registered using DMO API. Kindly get this created offline.
5103	Invalid Permanent Account Number	PAN Number entered is invalid.
5104	Invalid Merchant Alias Name	Merchant Alias name is invalid.
5105	Invalid Merchant Type	Merchant Type is incorrect. Kindly refer specifications.

5106	Invalid Merchant Address	Address of Merchant registered is incorrect.
5107	Invalid Merchant City	City of Merchant registered is incorrect
5108	Invalid Merchant State	State of Merchant registered is incorrect.
5109	Invalid SMS Notification	Kindly check specifications for correct value.
5110	Invalid Settlement Account Same As Parent	Kindly check the specifications for correct value.

5111	Invalid Merchant Pin Code	PinCode value entered is incorrect.
5112	Invalid Merchant Genre	Kindly refer the specifications for correct value.
5113	Invalid Channel	Pass the correct Channel value as provided by bank.

5114	Invalid Direct Push	Kindly refer the specifications for correct value.
5115	Received Pan Name from NSDL is blank	Check the PAN Number entered and retry again.
5116	Pan Number should be 10 digit length	Check the Pan Number entered.
5117	Received Account Name from ICORE is blank	Kindly check the Account Number entered and retry
5118	Received Account Name from IMPS is blank	Kindly check the Account Number entered and
5119	Virtual Address already exists	VPA used is already present in system. Use a new VPA for registration.
5301	StartDate Cannot be Greater than EndDate	Ensure that the validityStartDateTime is lesser than the validityEndDateTime
5302	Invalid Date Please Enter Date and Time in DD/MM/YYYY HH:MM:SS Format	

5304	Please Enter RefId when Update Status is Y	
5305	QR is already Used	

Security:

- API Key needs to be passed in every request in the header and merchant IP will also be required for IP whitelisting.
- API Key needs to be passed in the parameter name: apikey
- API request and response to Merchant is secured using advanced and agreed upon encryption algorithm agreed to maintain data confidentiality and integrity.
- API Gateway uses the standard authenticating and authorizing process for the incoming request from merchant and for maintaining the integrity and confidentiality we apply state of art Encryption/Decryption algorithm.

Encryption & Decryption Process:

For Encryption of a payload at Client's end.

```

encryptedKey =
Base64Encode(RSA/ECB/PKCS1Encryption(SesionKey,ICICIPubKey.cer)) Session key is nothing but randomly one
time generated string of length 16 (OR 32). encryptedData =
Base64Encode(AES/CBC/PKCS5Padding(Response,SessionKey))

```

1. Generate 16-digit random number (session key). Say RANDOMNO.
2. Encrypt RANDOMNO using RSA/ECB/PKCS1Padding and encode using Base64. Say ENCR_KEY.
3. Perform AES/CBC/PKCS5Padding encryption on request payload using RANDOMNO as key and iv-initialization vector. Say ENCR_DATA.
4. Now client may choose to send IV in request from one of below two options.
 - a. Send Base64 Encoded IV in "iv" tag. (Recommended Approach)
 - b. Send IV as a part of ENCR_DATA itself.

```
bytes[] iv = IV;
```

```
bytes[] cipherText = symmetrically encrypted Bytes (step3)
```

```
bytes[] concatB = iv + cipherText;
```

```
ENCR_DATA = B64Encode(concatB);
```

5. Now in the complete request, Client needs to send encrypted request in below format.

```
{  
  "requestId": "<request-id for tracking purpose>",  
  "service": "AccountCreation",  
  
  "encryptedKey": "<ENCR_KEY>",  
  "oaepHashingAlgorithm": "NONE",  
  "iv": "<IV>",  
  "encryptedData": "<ENCR_DATA>",  
  "clientInfo": "",  
  
  "optionalParam": ""  
}
```

For Decryption of a response at Client's end.

```
IV= getFirst16Bytes(Base64Decode(encryptedData))
```

SessionKey =

Base64Decode(RSA/ECB/PKCS1Decryption(encryptedKey,ClientPrivateKey.p12,)) Session key is nothing but randomly generated string of length 16 (OR 32) .

```
Response = Base64Decode (AES/CBC/PKCS5Padding Decryption(encryptedData,SessionKey, IV))
```

1. Get the IV- Base64 decode the encryptedData and get first 16 bytes and rest is encryptedResponse.

```
bytes[] IV= getFirst16Bytes(Base64Decode(encryptedData))
```

2. Decrypt encryptedKey using algo (RSA/ECB/PKCS1Padding) and Client's private key.

sessionKey =

```
B64Decode(RSA/ECB/PKCS1Decryption(encryptedKey,ClientPrivateKey.p12,))
```

3. Decrypt the response using algo AES/CBC/PKCS5Padding.

Response = Base64Decode

(AES/CBC/PKCS5Padding

Decryption(encryptedData,SessionKey, IV))

4. You need to skip first 16 bytes of response, as it contains IV.